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Section-B

Course : B .Tech CSE Core

Assignment: 1

❖ **Aim :**
To build a **dynamic To-Do List web application** using **HTML, CSS, JavaScript**, and a **database backed** to **add, edit, delete, and manage tasks permanently** with full CRUD functionality.

❖ **Tasks:**

- **Project Setup & Introduction** – Create project folders (frontend & backend) and link HTML, CSS, JS files.
- **UI Layout (HTML)** – Input field for task, optional description, priority dropdown, Add Task button, task list section, filter buttons (All/Active/Completed), and counters.
- **Styling (CSS)** – Centered , mobile-friendly layout, completed task styling, priority badges, button hover effects.
- **Frontend Interactivity (JS)** – Add, edit, delete, complete tasks; filter tasks; show counts; input validation.
- **Backend + Database** – Use Node.js + Express + SQLite (or other backend) to persist tasks.
- **REST API Implementation** – CRUD endpoints: GET, POST, PUT, PATCH, DELETE for tasks.
- **Bonus Features (Optional)** – Search, sort, due dates, local Storage fallback, JSON export, basic authentication.

❖ **Tech Stack Used:**
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js, Express
- **Database:** SQLite (or MySQL/other as per setup)
- **APIs:** RESTful APIs for CRUD operations

❖ Html Code :

```
<!doctype html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>My To-Do List</title>

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&displa
y=swap" rel="stylesheet">

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <main class="app-shell">

    <section class="todo-app">

      <!-- HERO -->
```

```
<div class="hero">
  <h1>My To-Do List</h1>
  <p class="subtitle">
  </p>
</div>

<!-- FORM -->
<section class="panel form-panel">

  <form id="taskForm" class="task-form">

    <div class="field-group title-group">
      <label for="taskTitle">Task title</label>

      <input
        id="taskTitle"
        name="title"
        type="text"
        maxlength="50"
        placeholder="Enter task title"
        required
      >

      <small class="hint">
        Required · maximum 50 characters
      </small>
    </div>
```

```
<div class="field-group">
  <label for="taskDescription">Description (optional)</label>

  <textarea
    id="taskDescription"
    name="description"
    rows="3"
    placeholder="Add more details about the task">
  </textarea>
</div>

<div class="field-row">

  <div class="field-group compact-field">
    <label for="taskPriority">Priority</label>

    <select id="taskPriority" name="priority">
      <option value="Low">Low</option>
      <option value="Medium" selected>Medium</option>
      <option value="High">High</option>
    </select>
  </div>

  <div class="field-group compact-field">
    <label for="sortTasks">Sort by</label>
```

```
      <select id="sortTasks" name="sortTasks">
        <option value="createdAt-desc">Newest first</option>
        <option value="createdAt-asc">Oldest first</option>
        <option value="priority-asc">Priority: High to Low</option>
        <option value="priority-desc">Priority: Low to High</option>
        <option value="title-asc">Title A-Z</option>
      </select>
    </div>

  </div>

  <div class="form-actions">

    <button type="submit" id="submitButton" class="primary-btn">
      Add Task
    </button>

    <button
      type="button"
      id="cancelEditButton"
      class="secondary-btn hidden">
      Cancel Edit
    </button>

  </div>

</form>
```

```
</section>

<!-- TOOLBAR -->
<section class="panel toolbar-panel">

  <div class="search-box">
    <label class="sr-only" for="searchTasks">Search tasks</label>

    <input
      id="searchTasks"
      type="search"
      placeholder="Search by title or description"
    >
  </div>

  <div class="filter-group" role="group" aria-label="Filter tasks">

    <button type="button" class="filter-btn active" data-filter="all">
      All
    </button>

    <button type="button" class="filter-btn" data-filter="active">
      Active
    </button>

    <button type="button" class="filter-btn" data-filter="completed">
```

```
      Completed
    </button>

  </div>

</section>

<!-- STATS -->
<section class="panel stats-panel">

  <div>
    <span class="stat-label">Completed / Total</span>
    <strong id="taskCounter">0 / 0</strong>
  </div>

  <div>
    <span class="stat-label">Active tasks</span>
    <strong id="activeCounter">0</strong>
  </div>

  <button type="button" id="exportButton" class="secondary-btn">
    Export JSON
  </button>

</section>

<!-- TASK LIST -->
```

```
    <section class="panel list-panel">

      <div id="statusMessage" class="status-message" aria-live="polite"></div>

      <ul id="taskList" class="task-list"></ul>

      <div id="emptyState" class="empty-state">
        <h2>No tasks found</h2>
        <p>Add your first task or adjust the filters/search.</p>
      </div>

    </section>

  </section>

</main>

<!-- TASK TEMPLATE -->
<template id="taskTemplate">

  <li class="task-item">

    <label class="checkbox-wrap">
      <input type="checkbox" class="task-checkbox">
      <span class="checkbox-ui"></span>
    </label>
```

```
  <div class="task-content">

    <div class="task-header-row">
      <h3 class="task-title"></h3>
      <span class="priority-badge"></span>
    </div>

    <p class="task-description"></p>

    <span class="task-date"></span>

  </div>

  <div class="task-actions">

    <button type="button" class="icon-btn edit-btn">
      Edit
    </button>

    <button type="button" class="icon-btn delete-btn">
      Delete
    </button>

  </div>

</li>
```

```
  </template>


  <script src="script.js"></script>


</body>
</html>
```

❖ CSS Code :

```
❖ :root{
❖     --primary:#6366f1;
❖     --primary-dark:#4f46e5;
❖     --success:#22c55e;
❖     --warning:#f59e0b;
❖     --danger:#ef4444;
❖     --text:#ffffff;
❖     --muted:#cbd5f5;
❖     --radius:14px;
❖ }
❖
❖ /* RESET */
❖
❖ *{
❖     box-sizing:border-box;
❖     margin:0;
❖     padding:0;
❖ }
❖
❖ /* BACKGROUND */
❖
❖ body{
❖     font-family:"Inter",sans-serif;
❖
❖     background:
❖     linear-gradient(rgba(0,0,0,0.45),rgba(0,0,0,0.45)),
❖     url("https://s3.amazonaws.com/eap04.easyagentpro.com-live/wp-
content/uploads/sites/165/2023/10/27125126/Discipline.jpg");
❖
❖     background-size:cover;
❖     background-position:center;
❖     background-repeat:no-repeat;
```

```
    background-attachment:fixed;

    min-height:100vh;
    color:white;
}

/* APP CONTAINER */

.app-shell{
    max-width:1100px;
    margin:auto;
    padding:40px 20px;
}

.todo-app{
    display:grid;
    gap:22px;
}

/* HERO */

.hero{
    text-align:center;
}

.hero h1{
    font-size:2.8rem;
    font-weight:700;
}

.subtitle{
    color:#e2e8f0;
}

/* GLASS PANELS */

.panel{
    background:rgba(255,255,255,0.08);
    backdrop-filter:blur(12px);
    border:1px solid rgba(255,255,255,0.2);
    border-radius:var(--radius);
    padding:22px;
    box-shadow:0 8px 30px rgba(0,0,0,0.3);
    color:white;
```

```
}

/* FORM */

.task-form{
    display:grid;
    gap:16px;
}

.field-group{
    display:grid;
    gap:6px;
}

input,textarea{
    background:rgba(255,255,255,0.15);
    border:1px solid rgba(255,255,255,0.3);
    border-radius:10px;
    padding:12px;
    color:white;
    font-size:14px;
}

input::placeholder,
textarea::placeholder{
    color:#e2e8f0;
}

input:focus,
textarea:focus{
    outline:none;
    border-color:var(--primary);
    box-shadow:0 0 0 2px rgba(99,102,241,0.4);
}

/* SELECT DROPDOWN */

select{
    background:rgba(255,255,255,0.15);
    border:1px solid rgba(255,255,255,0.3);
    border-radius:10px;
    padding:12px;
    color:black;
}
```

```
/* DROPDOWN OPTIONS */

option{
    color:black;
    background:white;
}

/* BUTTONS */

.primary-btn{
    background:linear-gradient(135deg,#6366f1,#8b5cf6);
    border:none;
    padding:12px 20px;
    border-radius:10px;
    color:white;
    font-weight:600;
    cursor:pointer;
    transition:0.25s;
}

.primary-btn:hover{
    transform:translateY(-2px);
    box-shadow:0 6px 18px rgba(99,102,241,0.5);
}

.secondary-btn{
    background:rgba(255,255,255,0.2);
    border:none;
    padding:12px 20px;
    border-radius:10px;
    color:white;
    cursor:pointer;
    transition:0.2s;
}

.secondary-btn:hover{
    background:rgba(255,255,255,0.35);
}

/* TOOLBAR */

.toolbar-panel{
    display:flex;
```

```
    justify-content:space-between;
    align-items:center;
    gap:14px;
    flex-wrap:wrap;
}

.search-box input{
    width:260px;
}

/* FILTER */

.filter-group{
    display:flex;
    gap:8px;
}

.filter-btn{
    background:rgba(255,255,255,0.15);
    border:none;
    padding:8px 16px;
    border-radius:20px;
    color:white;
    cursor:pointer;
}

.filter-btn.active{
    background:var(--primary);
}

/* STATS */

.stats-panel{
    display:flex;
    justify-content:space-between;
    align-items:center;
    flex-wrap:wrap;
}

.stat-label{
    font-size:12px;
    color:#cbd5f5;
}
```

```
/* TASK LIST */

.task-list{
    list-style:none;
    display:grid;
    gap:12px;
}

/* TASK ITEM */

.task-item{
    display:grid;
    grid-template-columns:auto 1fr auto;
    gap:16px;
    padding:16px;
    border-radius:12px;
    background:rgba(255,255,255,0.12);
    border:1px solid rgba(255,255,255,0.25);
    backdrop-filter:blur(10px);
    transition:0.2s;
}

.task-item:hover{
    transform:translateY(-3px);
    box-shadow:0 6px 18px rgba(0,0,0,0.4);
}

/* CHECKBOX */

.checkbox-ui{
    width:20px;
    height:20px;
    border-radius:50%;
    border:2px solid white;
    display:inline-block;
}

.task-checkbox{
    display:none;
}

.task-checkbox:checked + .checkbox-ui{
    background:var(--success);
    border-color:var(--success);
```

```
}

/* TASK CONTENT */

.task-title{
    font-size:16px;
    font-weight:600;
}

.task-description{
    margin:6px 0;
    color:#e2e8f0;
}

.task-date{
    font-size:12px;
    color:#cbd5f5;
}

.task-item.completed .task-title{
    text-decoration:line-through;
    opacity:0.6;
}

/* PRIORITY */

.priority-badge{
    font-size:12px;
    font-weight:600;
    padding:4px 10px;
    border-radius:20px;
}

.priority-low{
    background:#22c55e;
}

.priority-medium{
    background:#f59e0b;
}

.priority-high{
    background:#ef4444;
}
```

```
/* ACTION BUTTONS */

.task-actions{
    display:flex;
    gap:6px;
}

.icon-btn{
    border:none;
    background:rgba(255,255,255,0.2);
    padding:6px 10px;
    border-radius:8px;
    color:white;
    cursor:pointer;
}

.icon-btn:hover{
    background:rgba(255,255,255,0.35);
}

/* EMPTY */

.empty-state{
    text-align:center;
    color:#cbd5f5;
    padding:30px;
}

/* MOBILE */

@media(max-width:700px){

.task-item{
grid-template-columns:auto 1fr;
}

.task-actions{
grid-column:1/-1;
}

.toolbar-panel{
flex-direction:column;
align-items:flex-start;
```

```
}

}
```

- **JS Code :**

- **Script.js :**

```
const TITLE_LIMIT = 50;

const taskForm = document.getElementById("taskForm");
const taskTitle = document.getElementById("taskTitle");
const taskDescription = document.getElementById("taskDescription");
const taskPriority = document.getElementById("taskPriority");

const taskList = document.getElementById("taskList");
const emptyState = document.getElementById("emptyState");

const taskCounter = document.getElementById("taskCounter");
const activeCounter = document.getElementById("activeCounter");

const taskTemplate = document.getElementById("taskTemplate");

let tasks = [];

/* LOAD TASKS */

document.addEventListener("DOMContentLoaded", () => {

  const savedTasks = localStorage.getItem("tasks");

  if(savedTasks){
    tasks = JSON.parse(savedTasks);
  }

  renderTasks();
});

/* ADD TASK */

taskForm.addEventListener("submit", function(e){

  e.preventDefault();

  const title = taskTitle.value.trim();
```

```
  const description = taskDescription.value.trim();
  const priority = taskPriority.value;

  if(!title){
    alert("Task title required");
    return;
  }

  if(title.length > TITLE_LIMIT){
    alert("Title must be less than 50 characters");
    return;
  }

  const task = {
    id: Date.now(),
    title,
    description,
    priority,
    isDone:false,
    createdAt:new Date()
  };

  tasks.unshift(task);

  saveTasks();

  renderTasks();

  taskForm.reset();
});

/* RENDER TASKS */

function renderTasks(){

  taskList.innerHTML="";

  tasks.forEach(task=>{

    const fragment = taskTemplate.content.cloneNode(true);

    const taskItem = fragment.querySelector(".task-item");
    const checkbox = fragment.querySelector(".task-checkbox");
    const title = fragment.querySelector(".task-title");
```

```
    const description = fragment.querySelector(".task-description");
    const date = fragment.querySelector(".task-date");
    const badge = fragment.querySelector(".priority-badge");
    const deleteBtn = fragment.querySelector(".delete-btn");

    title.textContent = task.title;
    description.textContent = task.description || "No description";

    date.textContent = new Date(task.createdAt).toLocaleString();

    badge.textContent = task.priority;

    badge.classList.add(`priority-${task.priority.toLowerCase()}`);

    checkbox.checked = task.isDone;

    taskItem.classList.toggle("completed",task.isDone);

    checkbox.addEventListener("change",()=>{

      task.isDone = checkbox.checked;

      saveTasks();

      renderTasks();
    });

    deleteBtn.addEventListener("click",()=>{

      tasks = tasks.filter(t=>t.id!==task.id);

      saveTasks();

      renderTasks();
    });

    taskList.appendChild(fragment);
  });

  emptyState.style.display = tasks.length ? "none":"block";

  updateCounters();
}
```

```
/* SAVE TASKS */

function saveTasks(){

  localStorage.setItem("tasks",JSON.stringify(tasks));
}

/* UPDATE COUNTERS */

function updateCounters(){

  const total = tasks.length;

  const completed = tasks.filter(t=>t.isDone).length;

  taskCounter.textContent = `${completed} / ${total}`;

  activeCounter.textContent = total-completed;
}
```

❖ Package.json:

{ "name": "dynamic-todo-list-backend", "version": "1.0.0", "description": "Backend API for the Dynamic To-Do List assignment", "main": "server.js", "scripts": {

"start": "node server.js", "dev": "node server.js"

},"keywords": ["todo", "express", "sqlite"

],"license": "ISC", "dependencies": { "cors": "^2.8.6", "express": "^4.22.1", "sqlite3": "^5.1.7"

},"author": "", "type": "commonjs"

}

❖Server.js:

const express = require("express");

const cors = require("cors");

const path = require("path");

```
const sqlite3 = require("sqlite3").verbose();

const app = express();
const PORT = process.env.PORT || 4000;

/* DATABASE */

const dbPath = path.join(__dirname, "db.sqlite");
const db = new sqlite3.Database(dbPath);

/* FRONTEND */

const frontendPath = path.join(__dirname, "..", "frontend");

/* MIDDLEWARE */

app.use(cors());
app.use(express.json());
app.use(express.static(frontendPath));

/* CREATE TABLE */

db.serialize(() => {
  db.run(`
    CREATE TABLE IF NOT EXISTS tasks (
      id INTEGER PRIMARY KEY AUTOINCREMENT,
      title TEXT NOT NULL,
```

```
      description TEXT DEFAULT '',
      priority TEXT DEFAULT 'Medium',
      isDone INTEGER DEFAULT 0,
      createdAt DATETIME DEFAULT CURRENT_TIMESTAMP
    )
  `);
});

/* HELPERS */

function normalizeTask(row) {
  return {
    ...row,
    isDone: Boolean(row.isDone),
  };
}

function validateTitle(title) {
  return (
    typeof title === "string" &&
    title.trim().length > 0 &&
    title.trim().length <= 50
  );
}

function validatePriority(priority) {
  return ["Low", "Medium", "High"].includes(priority);
```

```
}

/* GET ALL TASKS */

app.get("/tasks", (req, res) => {

  db.all(
    "SELECT id,title,description,priority,isDone,createdAt FROM tasks",
    [],
    (error, rows) => {

      if (error) {
        return res.status(500).json({ message: "Failed to fetch tasks." });
      }

      const tasks = rows.map(normalizeTask);

      res.json(tasks);
    }
  );
});

/* CREATE TASK */

app.post("/tasks", (req, res) => {

  const { title, description = "", priority = "Medium" } = req.body;
```

```
if (!validateTitle(title)) {
  return res.status(400).json({
    message: "Title must be between 1 and 50 characters",
  });
}

if (!validatePriority(priority)) {
  return res.status(400).json({
    message: "Priority must be Low, Medium, or High",
  });
}

const query =
  "INSERT INTO tasks (title,description,priority) VALUES (?,?,?)";

db.run(
  query,
  [title.trim(), description.trim(), priority],
  function (error) {

    if (error) {
      return res.status(500).json({ message: "Failed to create task" });
    }

    db.get(
      "SELECT id,title,description,priority,isDone,createdAt FROM tasks WHERE id = ?",
```

```
        [this.lastID],
        (err, row) => {

          if (err) {
            return res.status(500).json({ message: "Task created but fetch failed" });
          }

          res.status(201).json(normalizeTask(row));
        }
      );
    }
  );
});

/* UPDATE TASK */

app.put("/tasks/:id", (req, res) => {

  const { id } = req.params;
  const { title, description = "", priority = "Medium" } = req.body;

  if (!validateTitle(title)) {
    return res.status(400).json({ message: "Invalid title" });
  }

  if (!validatePriority(priority)) {
    return res.status(400).json({ message: "Invalid priority" });
```

```
}

const query =
  "UPDATE tasks SET title=?,description=?,priority=? WHERE id=?";

db.run(
  query,
  [title.trim(), description.trim(), priority, id],
  function (error) {

    if (error) {
      return res.status(500).json({ message: "Failed to update task" });
    }

    if (this.changes === 0) {
      return res.status(404).json({ message: "Task not found" });
    }

    db.get(
      "SELECT id,title,description,priority,isDone,createdAt FROM tasks WHERE id=?",
      [id],
      (err, row) => {

        if (err) {
          return res.status(500).json({ message: "Task updated but fetch failed" });
        }
```

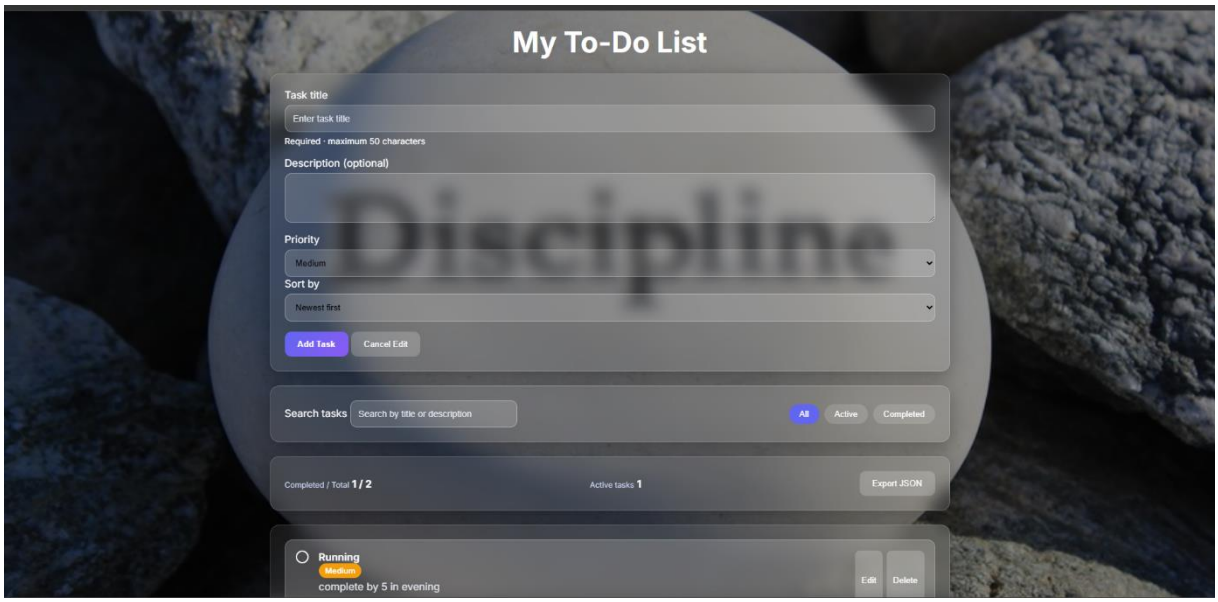
```
          res.json(normalizeTask(row));
        }
      );
    }
  );
});

/* TOGGLE STATUS */

app.patch("/tasks/:id/status", (req, res) => {

  const { id } = req.params;
  const { isDone } = req.body;

  if (typeof isDone !== "boolean") {
    return res.status(400).json({ message: "isDone must be boolean" });
  }

  db.run(
    "UPDATE tasks SET isDone=? WHERE id=?",
    [isDone ? 1 : 0, id],
    function (error) {

      if (error) {
        return res.status(500).json({ message: "Failed to update status" });
      }
```

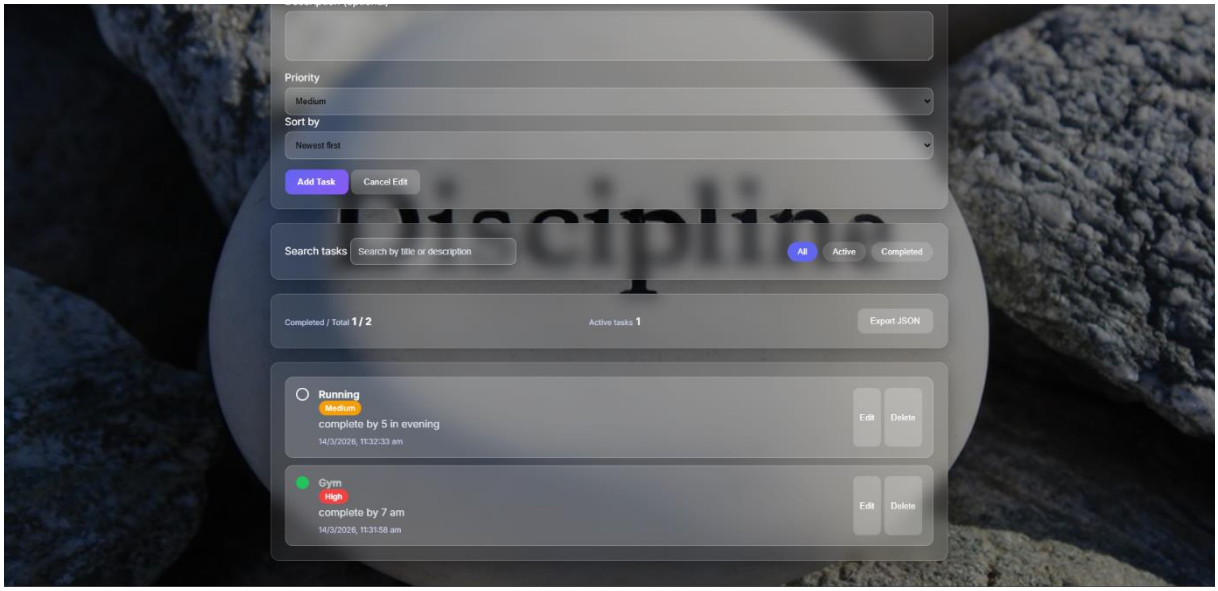
```
      if (this.changes === 0) {
        return res.status(404).json({ message: "Task not found" });
      }

      db.get(
        "SELECT id,title,description,priority,isDone,createdAt FROM tasks WHERE id=?",
        [id],
        (err, row) => {

          if (err) {
            return res.status(500).json({ message: "Status updated but fetch failed" });
          }

          res.json(normalizeTask(row));
        }
      );
    }
  );
});

/* DELETE TASK */

app.delete("/tasks/:id", (req, res) => {

  const { id } = req.params;

  db.run(
```

```
    "DELETE FROM tasks WHERE id=?",
    [id],
    function (error) {

      if (error) {
        return res.status(500).json({ message: "Failed to delete task" });
      }

      if (this.changes === 0) {
        return res.status(404).json({ message: "Task not found" });
      }

      res.status(204).send();
    }
  );
});

/* FRONTEND ROUTE */

app.get("*", (req, res) => {
  res.sendFile(path.join(frontendPath, "index.html"));
});

/* START SERVER */

app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
```

```
});

/* CLOSE DATABASE */

process.on("SIGINT", () => {
  db.close(() => {
    process.exit(0);
  });
});
```

❖ Output:




Priority
Medium
Sort by
Newest first
Add Task
Cancel Edit
Search tasks
Search by title or description
All
Active
Completed
Completed / Total 1 / 2
Active tasks 1
Export JSON
Running
Medium
complete by 5 in evening
14/3/2026, 11:32:33 am
Edit
Delete
Gym
High
complete by 7 am
14/3/2026, 11:31:58 am
Edit
Delete